ONTOLOGY

Based solution using Protégé

ABSTRACT

Informative and investigative report based on problem domain, by providing technical evidence using protégé as an ontology based solution.

CONTENTS

INTRODUCTION OF ONTOLOGY

Ontology is based on solution related domain problems, which helps to rectify the data properties, Object properties and relations in a conceptual way of understanding. The terminology of Ontology is called the concept which we use as a method to sort out the problem. According to Barry Smith, who is known as the philosopher has explained this relating "Ontology as the branch of philosophy which is the science of what the kinds and the structures of objects, Properties, events, processes and relations in every area of reality." (Smith, 2012).

Ontology is a widely used technic in the form of knowledge based program in different kinds of industry. The representational of an ontology provides a clear understanding of classes, relations, functions and objects. This helps in understanding the concepts that is created for a domain, showing the relationship between each and every entities. It can be based on industries such a food production company, Hospital, Schools, Universities and many other organizations. This can also be for the things like foods, medicines or other objects which can be listed as classes in relation to their entities.

Therefore I've chosen the Concept of a Hospital based domain which connects the staffs and the departments of the hospital network. The ontology concept that I have selected is provide with the protégé, technical evidence in understanding the conceptualization. Basically this methodology according to computer science is known as the "catalogue of types of things that are assumed to exist in a domain of interest." (k.Breitman, 2004) . This is also as the gathering of semantics of data, where as you can gather a group of data according to the way of machine understanding.

As following my investigation in the topic ontology I was able to find a easier way to provide the explanation in my report for the topic ontology. Below I have shown the steps that is used in the way of building an ontology.

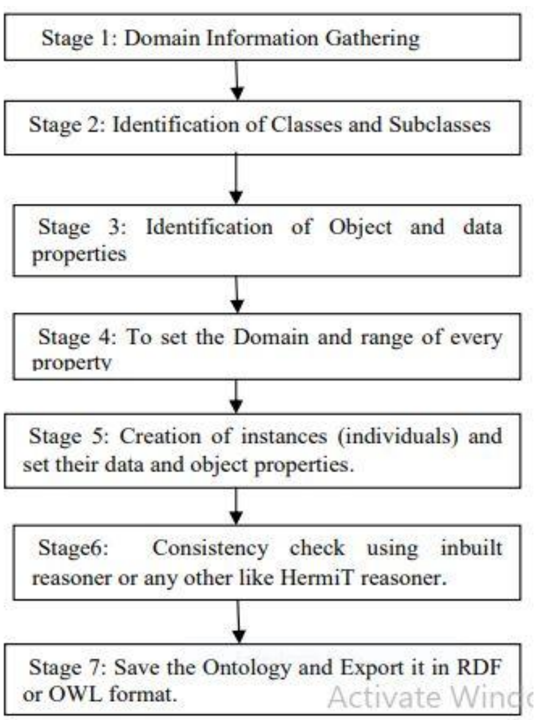

Figure 1: Steps for Ontology Building, *(Ritika Bansal, 2014)*

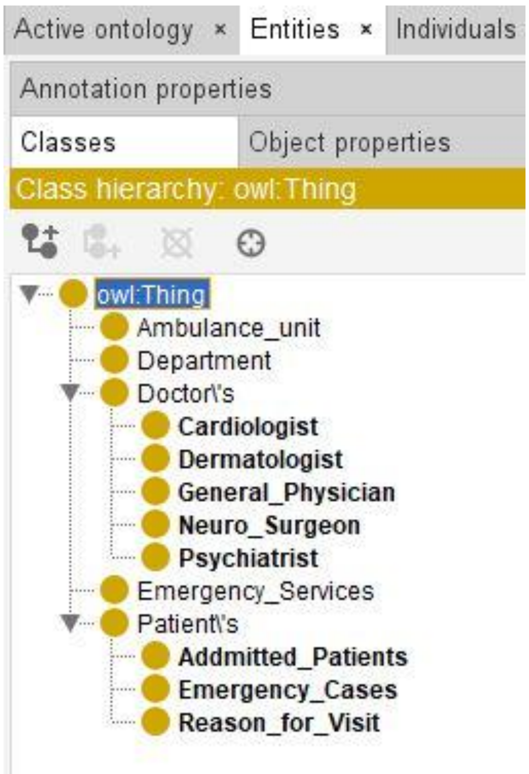

Figure 2: Class Hierarchy from OWL thing

This is my class hierarchy, which I have presented here with the help of Protégé. This is also known as the class hierarchy of the OWL thing. I have selected the hospital domain as my suitable concept for this ontology project. The above image shows the classes that I have taken in presenting my ontology report. The classes carries the sub classes whereas they are connected with relation to their connections. My classes are Doctor's, Department, Ambulance Unit, Emergency Services and patient's. These classes holds sub classes which they are related to. For example with relating to my class hierarchy of doctor's the sub classes are (Cardiologist, Dermatologist, General Physician, Neuro Surgeon and Psychiatrist. These are the sub classes where the doctors are specialized in the field of their work. So the relation between Class doctor's and its subclasses is that the Field of the doctor's they are specialized in.

But also some of my classes does not hold or categorize into subclasses that is because it has separate individuals which are not subclasses. Those individuals classifies the specification of its class. Therefore in my class hierarchy the classes' ambulance unit, Department and Emergency services has its individuals instead of the subclasses. Whereas ambulance unit has the individual which involves as the ambulance services. And there after the class department as its individuals as it represents the each department in the hospital. The departments are (cardiology department, Dermatology department, General ward, Neurology department, Psychology department).

The Class Patient's has its own subclasses which it specifies the relation in between the class and sub classes. The subclasses are (Admitted patient's, Emergency cases, Reason for visit).Theses are the sub classes which relates the other operational activities in the hospital industry.

ONTOLOGY CONSTRUCTION IN THE CONCEPT OF ER DIAGRAM

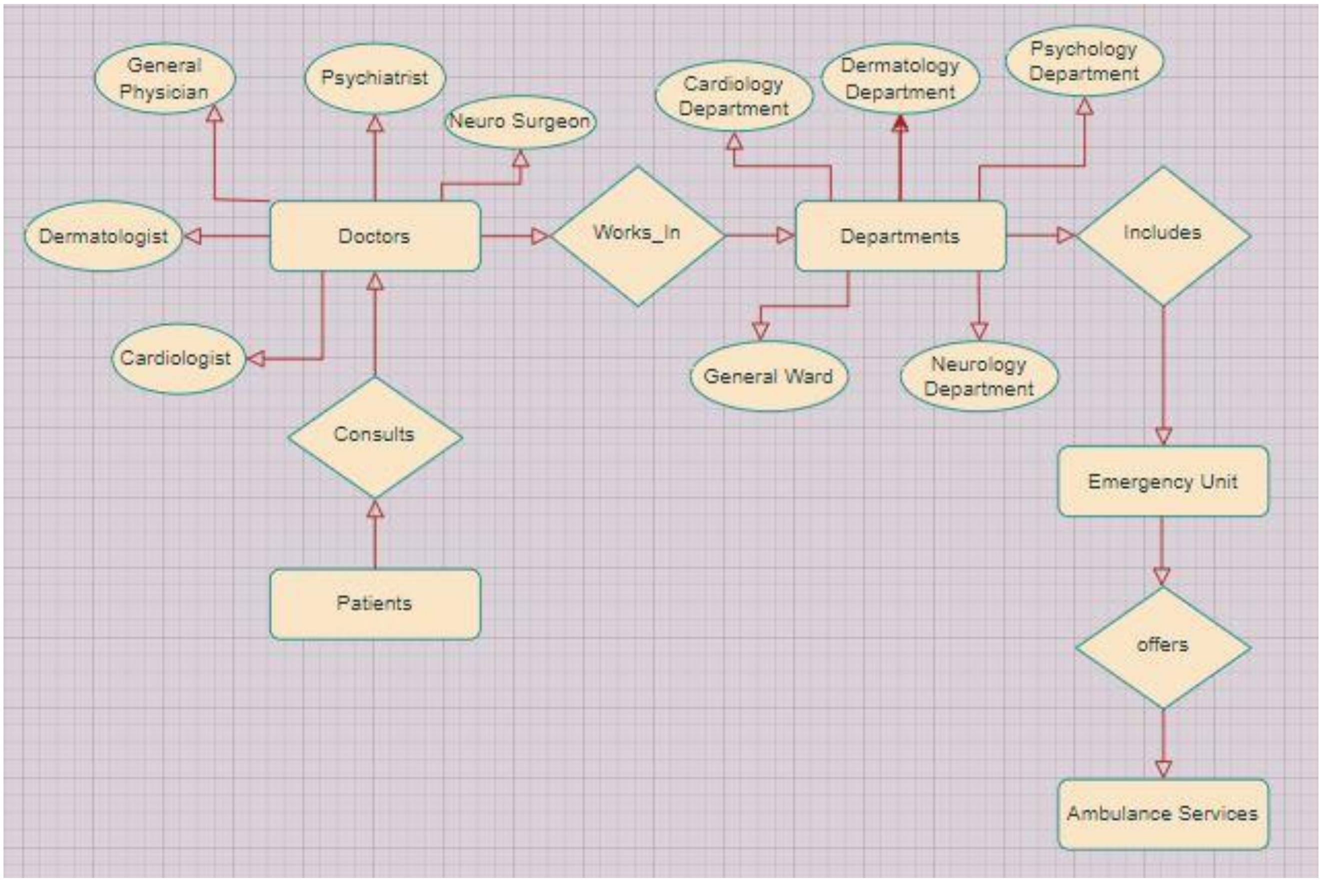

Figure 3: Ontology Construction in the concept of ER Diagram

The above shown image is based on the ontology construction of my selected domain. This is known as the ERD (Entity Relation Diagram), which presents my classes, sub classes and its individuals. Whereas this is shown in the format of entities and attributes according to the selected ontology. The diamond shape represents the relationship between the classes in my ontology and the circle shape images represents the subclasses which also can be said as (attributes).

Ontology metrics:

Metrics

Axiom	207
Logical axiom count	160
Declaration axioms count	47
Class count	13
Object property count	4
Data property count	8
Individual count	22
Annotation Property count	0

Figure 4: Ontology Metrics

EXPLANATION ON:

OBJECT PROPERTIES

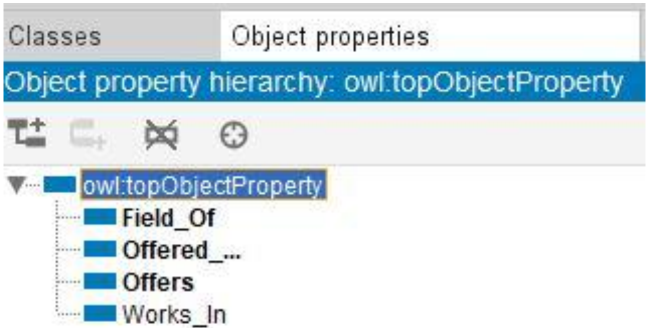

Figure 5: Object Property Hierarchy

The image above represents the object property hierarchy of my ontology. This specifies the object properties of the classes that are in relation to its subclasses. Object property helps to identify the concept and the helps identifying the required information in order to sort the problem in the domain. As per the journal that I have found throughout my investigation about ontology, I was able to find out a proper explanation for object properties. “Various object properties which describes the relationship between two instances or two individuals of the classes have been set.” (Ritika Bansal, 2014)

- **Ambulance services ------- (offered) ------Emergency Unit**

This shows the object property between the classes’ ambulance service and emergency unit is (offered).Which means that the ambulance service is offered by the emergency unit in the hospital.

- **Emergency unit ----------(Offers)----------Ambulance Services**

This shows the object property between the classes’ emergency unit and ambulance services (Offers).Which means that the Emergency unit offers the ambulance services in the hospital.

And then the relationship between the classes Doctor’s and Departments is classified with the term (Field of) and (works in).

DATA PROPERTIES

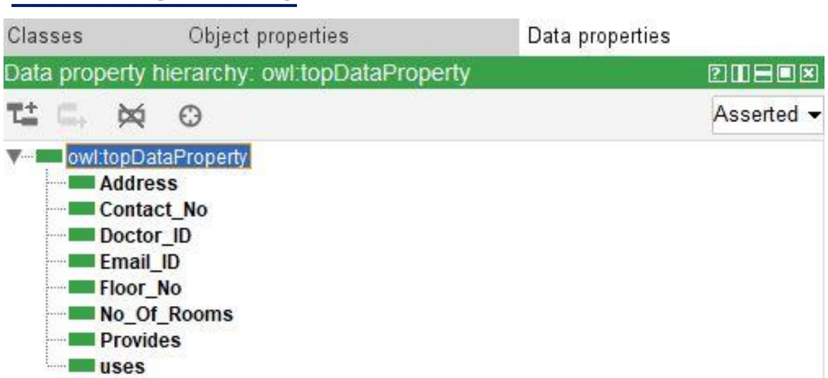

Figure 6: Data Property hierarchy

The image shown above is the data property hierarchy of my ontology. This data property hierarchy identifies the data that is/will be stored under this domain. It is simply said that the domain consist of data in the hospital hierarchy. As per my investigation the simple explanation for data properties is “data properties are number of operations and principle used.” (Ritika Bansal, 2014)

This classifies the data properties of each class:

(Address, Contact no, Doctor ID, Email ID) are the data properties of the class Doctor’s.

(Floor no and No of rooms) are the data properties of the class Department.

(Provides) this is the data property of the class Emergency Services.

(Uses) this is the data property of the class Ambulance Unit.

INDIVIDUALS

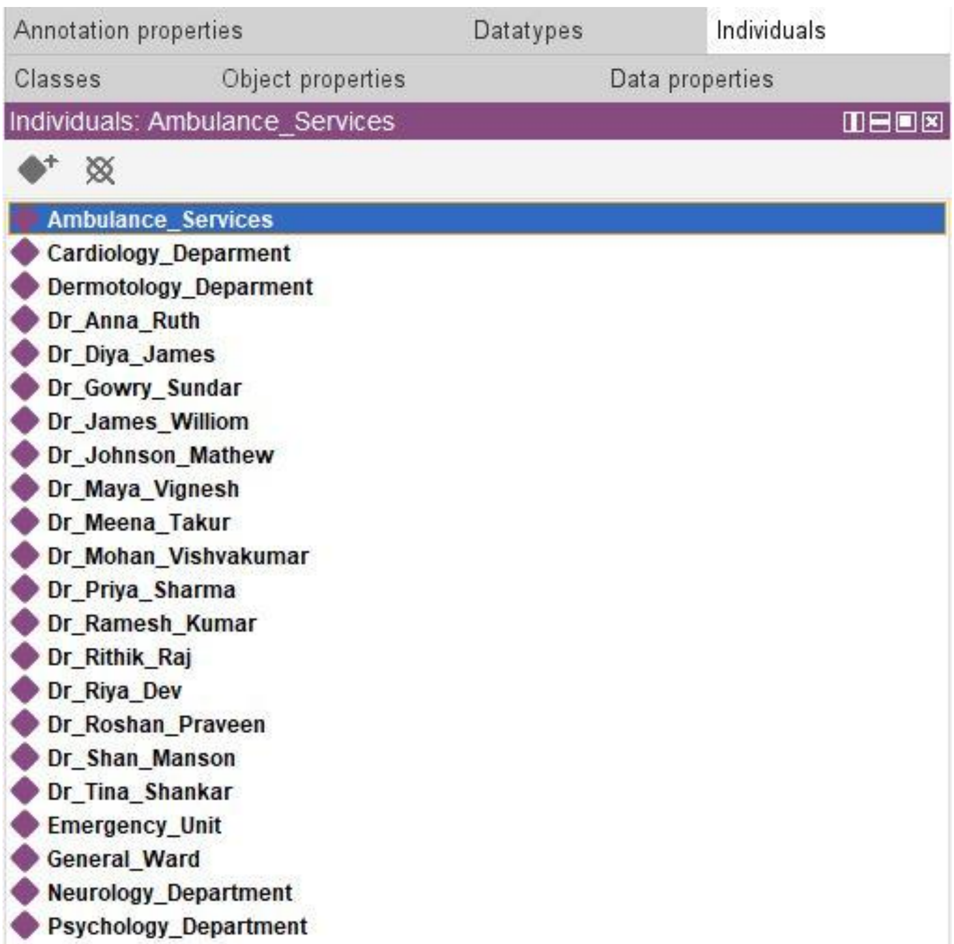

Figure 7: Individuals of the Classes--subclasses

“Individuals act as an instance for classes and subclasses. Through individuals (instance), relationships are formed between all the entities of the respective ontology.” (Ritika Bansal, 2014) This is a brief explanation given for the concept individuals as in the journal published by the IJEAT (International Journal of Engineering and Advance Technology). I have stated the list of individuals in the above image of my ontology for the hospital domain. These individuals are connected to its class and sub classes of the ontology.

ONTO GRAPH PRESENTATION

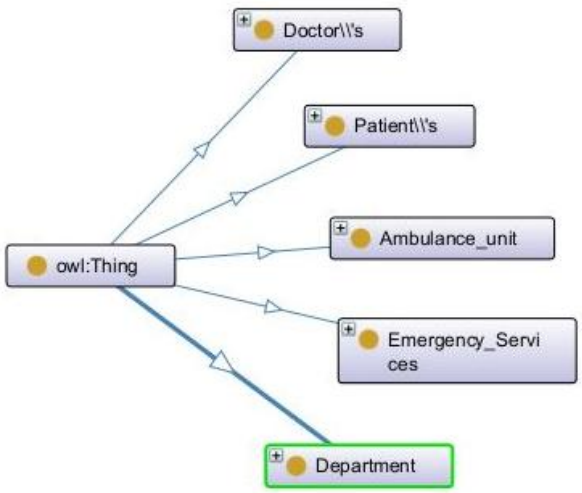

Figure 8: ONTO graph presentation of the main OWL thing of my ontology

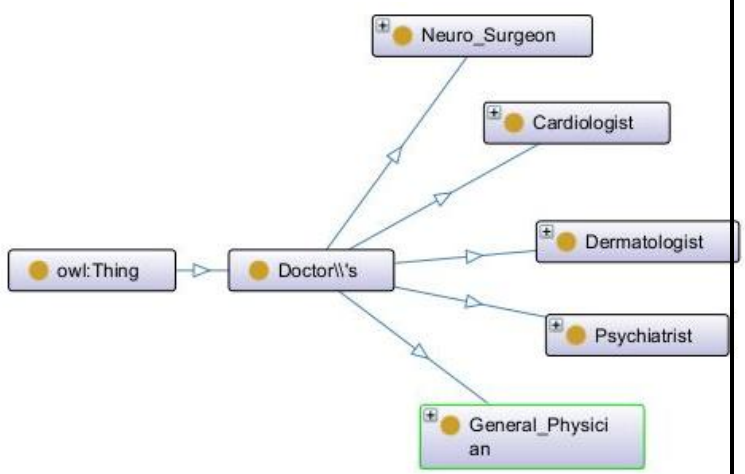

Figure 9: ONTO graph presentation of the class Doctor's

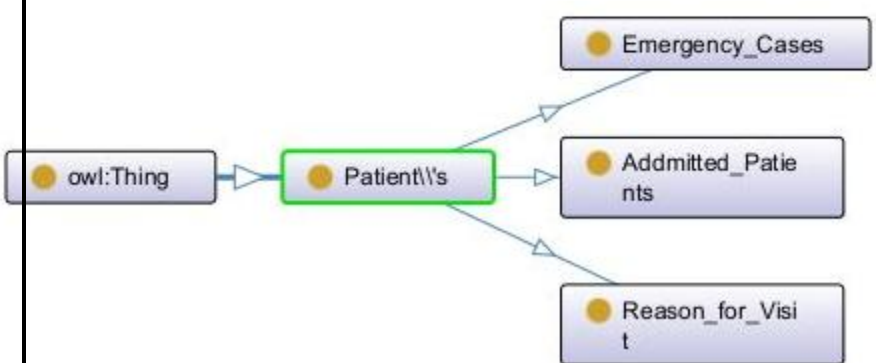

Figure 10: ONTO graph presentation of the class Patient's

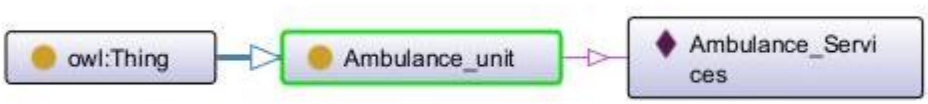

Figure 11: ONTO graph presentation of the class Ambulance Unit

Figure 12: ONTO graph presentation of the class Emergency Services

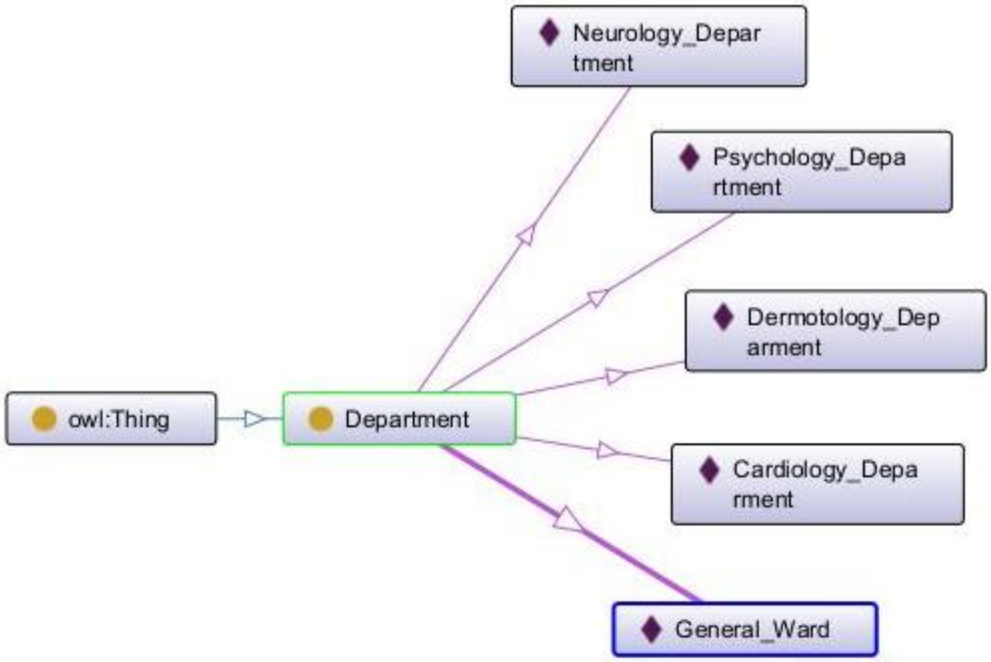

Figure 13: ONTO graph presentation of the class Department

The ONTO graph presentation shows the Classes and sub classes of my ontology, I was able to create this concept of the domain I have chosen with the help of protégé. ONTO graph will help the user understand about the ontology of OWL thing.

PART B: DISCUSSION ON AI IN SOCIETY

- **Could AI achieve human level intelligence or become conscious?**

In this fast moving world, Artificial Intelligence is playing a major role in human day to day life. It is said that the AI will over rule the world in future taking over the human intelligence. Technology is developing day by day in the co-operate world. For each and every day to day activities we as humans depend on all the electronic devices which is said to be the Artificial Intelligence. From mobile phone to television and clocks to electronic watches, Personal computers, Electronic Tablets, headphones and many other wireless and portable devices are the phase of Artificial Intelligence. All these electronic devices provides the outcome according to the human instructions.

From children's to adults are dependent on all of these electronic devices. From doing school home work to university Assingment, to cleaning house hold and cooking in the kitchen everything is electronically invented. The AI is aiming to achieve the human level intelligence such as "the ability to sense, understand, reason, learn, and cut in dynamic environments." (Leung, 2020)

Humans are reason for the development of the Artificial Intelligence, as we ourselves give more importance to the technology instead of the human interaction. Human have started interacting machines which is now known as the human computer interaction. The evolution of Artificial Intelligence started with smart phone to human designed robotics. Robotics are playing major role in all the industries such as hospitals, production and manufacturing companies, hotels, schools and universities. Now a days robots are replacing humans in all the field of work. This is where the point that needs to be discussed whether AI could achieve human level intelligence or become conscious. But the answer is no according to my argument. "The strong AI must be able to demonstrate sentience, emotional intelligence, imagination, effective command of other machines or robots and self-referring and qualities." (Leung, 2020) But machines cannot replace humans in senses and emotions. This is where AI fails to achieve the human level intelligence as they will not be able to fulfill the human level satisfaction. Machines makes error which can also be the human error and it is not proven that machines don't make mistakes. It is because there can be technical break downs in which the AI fails to reach the target.

"The Tala mind thesis accepts the objection by some AI skeptics that a system which is not aware of what it is doing and does not have some awareness of itself cannot be considered to have human level intelligence." (Jackson, 2018)

Therefore, there is no possibilities for the AI to achieve the human level intelligence. It is proven that AI is just a machine which just runs and performs its task accordingly programmed and constructed by human commands. There are arguments based on AI over ruling and replacing humans but it is not proven technically with evidence. In future there are chances of AI playing major role in each part of human life as it is already involving in human survival. As the technology is proving itself to be the best and developing in all forms and all part of the world. But there will be a major disconclussion for AI taking part the human revolution. "In the last 10 years, AI has been seen successes in fields such as natural language processing, Computer Vision, Speech Recognition, Robotics and Autonomous systems." (Leung, 2020)

As my conclusion, I would to conclude on the argument furter proceeding in my points saying that AI will not/ will ot be able to achieve the human level intelligence or become concious according to my investigation on this topic. But it is unpredictable to say that this can be achievable in future as the techology dominates the human survival.

❖ APPENDIX:

Assingment CW01.rdf

References

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